Composing an Origins-of-Life Simulation Campaign in Vivarium

Companion implementation dossier | 11 October 2026 | Scope: alkaline, serpentinization-driven hydrothermal vents

Purpose. Identify executable computational engines, numerical methods, chemical and biological databases, interoperability standards, and a staged integration strategy. This is a design and resource-selection document, not a claim that existing software already simulates abiogenesis end-to-end.

Core principle: keep geology, aqueous chemistry, transport, mineral surfaces, protometabolism, compartment dynamics, polymers, and population evolution as replaceable models coupled by explicit conserved quantities.

1 | Executive architecture

Scale / state	Recommended initial engine	Coupling output
Rock-fluid thermodynamics	PHREEQC or Reaktoro	aqueous species, pH, saturation indices, mineral extents
Flow, temperature, solute transport	FiPy / custom spatial-transport; OpenFOAM if needed	velocity, T, concentration fields
Aqueous / surface reactions	Cantera, SciPy ODEs, custom kinetic laws	species fluxes, heat, charge
Thermodynamic screening	eQuilibrator + curated reaction thermochemistry	reaction Gibbs energies, feasibility bounds
Chemical networks	Tellurium/RoadRunner, COPASI, GillesPy2	reaction-network state and event histories
Membranes / pores / polymers	Custom mesoscopic models; OpenMM/LAMMPS for calibration	permeability, assembly and adsorption laws
Protocell agents / selection	Vivarium process-bigraph + agent-based processes	birth, division, exchange, lineage and fitness

The recommended **first integrated composite** is a 1D/2D porous vent wall with externally prescribed fluid boundary conditions, thermodynamic speciation, reactive solute transport, a minimal Fe/NiS-catalyzed CO2-reduction network, and explicit mass/charge accounting. Serpentinization can initially enter as a boundary flux and later become a coupled mineral-kinetics module.

2 | Stage-by-stage resource map

Stage 0 - Environmental priors and Hadean scenarios

Scientific model: Use InterRidge vent metadata, Lost City field studies, seawater reconstructions and mineralogical priors to create ensembles of initial/boundary conditions. Reconstructing Hadean seawater is uncertain: retain alternative ocean pH, salinity, DIC, Fe, sulfur and phosphorus scenarios.

Computational resources: PANGAEA/InterRidge; EarthChem; published Lost City chemistry; PHREEQC thermodynamic databases.

Methods: Sample correlated environmental scenarios; distinguish directly measured modern analog values from inferred ancient parameters.

Vivarium contract: Outputs: T, P, flow, pH, DIC, H2, dissolved metals, mineral assemblage and uncertainty distributions.

Stage 1 - Serpentinization and geochemical disequilibrium

Scientific model: Model water-rock reaction, mineral dissolution/precipitation, Fe redox, hydrogen production, alkalinity and heat. Start with calibrated source terms; upgrade to kinetic mineral transformations.

Computational resources: PHREEQC (speciation, equilibrium, kinetic reactions); Reaktoro (reactive chemical equilibrium/kinetics); CrunchFlow/TOUGHREACT for larger reactive-transport applications.

Methods: Equilibrium speciation plus kinetic rate laws dM/dt = k A f(T,pH,activities); use reaction affinity and mineral surface area rather than fixed H2 yield.

Vivarium contract: Inputs: water and rock; outputs: mineral inventory, H2 source, alkalinity, ionic composition, enthalpy.

Stage 2 - Chimney growth and pore-scale flow

Scientific model: Simulate mixing of alkaline fluid and ocean, FeS/carbonate precipitation, changing porosity and permeability, temperature gradients and residence-time distributions.

Computational resources: FiPy finite-volume PDEs; OpenFOAM for resolved CFD; FEniCSx for FEM; pore-network modeling via PoreSpy/OpenPNM.

Methods: Advection-diffusion-reaction PDE; Darcy flow at continuum scale; optional Navier-Stokes for pore-scale geometry. Couple porosity to precipitation and permeability.

Vivarium contract: Outputs: pore geometry, flow, residence time, concentration/temperature fields.

Stage 3 - Electrochemical gradients and mineral catalysis

Scientific model: Compute H+ gradients, membrane/mineral interface potentials, ion transport and electron-transfer-limited CO2 reduction. A pH difference is not itself proof of extractable work.

Computational resources: Nernst-Planck-Poisson or electroneutral approximations with FiPy/FEniCSx; PHREEQC for aqueous speciation; Cantera/custom heterogeneous kinetics.

Methods: Flux $J_i = -D_i \nabla(c_i) - z_i u_i F c_i \nabla(\phi) + v c_i$; reaction-current coupling must obey charge conservation.

Vivarium contract: Outputs: electrochemical potentials, electron flux, reaction affinities, catalytic rates.

Stage 4 - Carbon fixation and protometabolism

Scientific model: Build a curated reaction graph from formate, acetate, pyruvate, glyoxylate and organic acids. Include measured nonenzymatic routes, reversible reactions and competing sinks.

Computational resources: SBML + libRoadRunner/Tellurium or COPASI; SciPy solve_ivp; Cantera for suitable reaction kinetics; eQuilibrator for thermodynamic checks.

Methods: ODE $dC/dt = S\, v(C,T,pH,catalyst) + transport$; thermodynamic consistency and element balance checked before fitting kinetics.

Vivarium contract: Outputs: carbon conversion, yields, network persistence, energy dissipation and selectivity.

Stage 5 - Concentration, adsorption and polymer chemistry

Scientific model: Represent thermophoresis, diffusion, convection, mineral adsorption, oligomerization and hydrolysis. Use explicit size distributions for polymers when tractable.

Computational resources: FiPy; custom finite-volume transport; GillesPy2/SSA for low copy numbers; OpenMM or LAMMPS for molecular-scale calibration only.

Methods: Size-dependent diffusion, adsorption/desorption, temperature-dependent polymerization and degradation; compare residence time against synthesis timescale.

Vivarium contract: Outputs: enrichment factors, polymer length distributions, survival of activated precursors.

Stage 6 - Amphiphiles, mineral pores and protocells

Scientific model: Model amphiphile supply, aggregation, mixed-membrane stability, permeability, osmotic balance and division; compare mineral-confined and free vesicles.

Computational resources: Custom coarse-grained membrane kinetics; Smoldyn/ReaDDy for spatial stochastic reactions; OpenMM/LAMMPS for targeted membrane parameter studies.

Methods: Compartment agents exchange species through flux laws; include leakage, ion partitioning, membrane area, tension and energy constraints.

Vivarium contract: Outputs: viable compartment count, chemical retention, membrane growth and stability.

Stage 7 - Catalytic polymers, replication and heredity

Scientific model: Introduce polymer sequence or class-based catalysis, template copying, error rates, degradation and genotype-phenotype coupling. Do not presume a viable RNA replication mechanism.

Computational resources: GillesPy2/StochPy for SSA; rule-based chemistry (BioNetGen/NFsim) for combinatorics; custom sequence-level agent dynamics.

Methods: Explicit replication fidelity, substrate activation, chemical costs and competition; separate catalyst discovery from inheritance.

Vivarium contract: Outputs: lineage persistence, information retention, evolvability and replication thresholds.

Stage 8 - Ecosystems, autonomy and pre-LUCA transitions

Scientific model: Couple heterogeneous compartments, cross-feeding, migration, environmental fluctuations, active transport and selection for independence from vent gradients.

Computational resources: Vivarium multi-agent composites; COBRApy for later genome-scale metabolic benchmarks; evolutionary algorithms / Bayesian calibration.

Methods: Agent dynamics + shared spatial fields; evaluate environmental energy dependence, ecological stability and population-level selection.

Vivarium contract: Outputs: autonomous lineages, metabolic diversity, cross-feeding and robustness.

3 | Software selection: what to use, and what not to assume

Resource	Best role	Caution / implementation note
PHREEQC	Aqueous speciation, saturation, mineral reactions	Thermodynamic database choice and activity model matter strongly at high ionic strength.
Reaktoro	Python-accessible equilibrium and reactive chemistry	Check supported thermodynamic datasets and kinetic formulations before committing.
FiPy	Rapid 1D/2D finite-volume transport	Coupling electrostatics, flow and evolving pore geometry requires development.
OpenFOAM / FEniCSx	Resolved flow / multiphysics FEM	Higher implementation and compute cost; defer until reduced models fail.
Cantera	Well-specified chemical kinetics and reactor networks	No prebuilt validated prebiotic vent reaction mechanism is supplied.
Tellurium / COPASI	SBML kinetic protometabolic networks	Kinetic constants for abiotic networks must be curated or inferred.
eQuilibrator	Reaction thermodynamics, screening	Biochemical standard-state assumptions may not suit hydrothermal brines.
Smoldyn / ReaDDy	Particle-based reaction-diffusion	Mesoscopic spatial models are not mineral-surface quantum chemistry.
OpenMM / LAMMPS	Focused atomistic/coarse-grained studies	Force-field validity and accessible timescales are major constraints.
COBRApy	Metabolic network analysis and late-stage benchmarks	Steady-state FBA should not replace prebiotic reaction kinetics.

4 | Vivarium integration contracts

Represent each model as a process with declared input/output stores, units, spatial support, step size, solver state and provenance. Suggested shared schema: **environment** (T, P, flow, potential); **aqueous** (species amount by voxel); **minerals** (phase abundance, surface area, porosity); **interfaces** (area, permeability, surface charge); **compartments** (volume, membrane, internal species); **lineages** (catalysts, sequences, ancestry).

Contract	Invariant / test
Chemical species	Canonical IDs (formula + charge + phase + protonation); use mol amounts internally, derive concentrations from volumes.
Mass / elements	Elemental inventory residuals tracked for C, H, O, N, P, S, Fe, Ni and others.
Charge / electrons	Explicit electroneutrality or Poisson closure; track redox and electron bookkeeping.
Energy	Thermodynamic directionality; log external work, heat and chemical free-energy flow.
Spatial transfer	Fluxes conservative across neighboring cells and compartment boundaries.

Contract	Invariant / test
Time integration	Operator splitting with convergence studies; avoid hidden double-updates to shared state.
Reproducibility	Version solver, database, parameter source, random seed and scenario identifier.

Illustrative update schedule: slow rock/mineral changes (minutes to days); transport and chemistry (milliseconds to hours, adaptively); membrane and agent changes (seconds to hours); evolution (many generations). This is a proposed multirate schedule, not an empirically established universal timescale.

5 | Model hierarchy and computational cost

Tier A - reduced screening: 0D/1D compartments, fixed geometry, prescribed vent source, 10-50 species and simple kinetic network. Thousands of parameter sweeps. **Tier B - mechanistic spatial:** 2D finite-volume fields, dynamic mineral surfaces, adsorption, electrochemical transport and explicit pore populations. Hundreds of runs. **Tier C - focused calibration:** pore-scale CFD, molecular dynamics or reactive mineral modeling to derive closure laws for Tiers A/B. **Tier D - evolution:** thousands of compartment agents coupled to coarse spatial fields. Use emulators for expensive chemistry where validated.

6 | Six computational campaigns and acceptance gates

I. Vent reactor

Couple geochemical source, transport, mineral precipitation, pH/redox gradients and CO2 reduction. **Gate:** Reproduce qualitative trends and published experimental regimes, including negative controls; conserve mass and charge.

II. Sustained protometabolism

Expand to acetate/pyruvate and TCA-like fragments, with thermodynamic and kinetic constraints. **Gate:** Identify parameter domains supporting nonzero sustained carbon throughput without imposed product accumulation.

III. Concentration and protocells

Add adsorption, thermophoresis, amphiphiles and permeability. **Gate:** Show measurable enrichment and retention, and map when hydrolysis/leakage defeat synthesis.

IV. Energy autonomy

Introduce catalytic membranes, ion selectivity and active transport as hypothetical innovations. **Gate:** Quantify reduced dependence on environmental gradients; compare energetic cost of membrane maintenance.

V. Heredity and evolution

Add copying fidelity, catalytic polymers and lineage selection. **Gate:** Demonstrate inheritance and adaptation under chemically feasible resource budgets.

VI. Ecological emergence

Cross-feeding and fluctuating vents; alternative shallow/terrestrial settings. **Gate:** Test robustness and distinguish vent-specific predictions from generic nonequilibrium chemistry.

7 | Calibration, uncertainty, inference and validation

Keep **measured**, **literature-derived**, **estimated** and **hypothetical** parameters distinct. Use hierarchical priors across mineral types and experimental setups; perform global sensitivity screening (Morris/Sobol), Bayesian calibration (PyMC/NumPyro), and posterior predictive checks against independent experimental conditions. Prefer profile likelihood or identifiability tests before fitting large kinetic networks. Experimental failure cases are mandatory validation targets.

Recommended metrics: H2 consumption, dissolved inorganic carbon fixation, product selectivity, carbon mass balance, charge balance, energy dissipation, pore residence-time distribution, enrichment ratio, membrane retention, network persistence, division rate and lineage survival. Record raw time series plus aggregated observables.

8 | Implementation roadmap (first 12 months)

Window	Deliverables
Months 1-2	Data registry; species and units schema; 0D vent/ocean mixer; PHREEQC/Reaktoro proof-of-concept; benchmark notebooks.
Months 3-4	1D FiPy transport + geochemical speciation + simple mineral precipitation; regression tests and parameter sweeps.
Months 5-6	CO2-to-formate/acetate kinetic process, experimental comparison, provenance-rich model cards; first integrated demonstration.
Months 7-9	Dynamic pore/adsorption/concentration models; compare alternative physical closures and gradient mechanisms.
Months 10-12	Compartment agents, amphiphile exchange and scenario comparisons; reproducible Workbench demonstration and manuscript-ready figures.

9 | Data and model provenance protocol

Each module should ship with a machine-readable **model card**: source paper/DOI; extracted figure/table; raw dataset URL; experimental system; units; temperature/pressure/pH range; fitted formula; parameter uncertainty; applicability bounds; solver/version; validation tests; and alternative parameterizations. Avoid transferring a rate constant between pH, pressure or mineral regimes without an explicit extrapolation flag.

Each simulation run should export: scenario YAML, parameter posterior/sample, code commit, solver/database versions, composite graph, random seeds, provenance manifest and conservation/validation diagnostics. Prefer SBML for kinetic networks, COMBINE/OMEX where practical for model packaging, and SED-ML for standardized simulation tasks. Use a separate spatial schema for meshes and compartment topology.

10 | Resource directory and primary computational references

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https://readdy.github.io/

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[18] BioNetGen. Faeder, Blinov & Hlavacek (2009), Rule-based modeling of biochemical systems with BioNetGen.
https://bionetgen.org/

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https://www.pymc.io/

[20] SBML. Keating et al. (2020), SBML Level 3: an extensible format for the exchange and reuse of biological models, Molecular Systems Biology.
https://sbml.org/

[21] SED-ML. Waltemath et al. (2011), Reproducible computational biology experiments with SED-ML, BMC Systems Biology.
https://sed-ml.org/

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https://combinearchive.org/

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https://www.usgs.gov/software/phreeqc-version-3

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11 | Key design decisions requiring early validation

Geochemical consistency: PHREEQC/Reaktoro may predict different results depending on database and activity model; establish a fixed reference suite. **Redox accounting:** separate chemical affinity from kinetically accessible electron transfer. **Surface chemistry:** do not equate mineral presence with catalytic activity; curate catalyst identity, area, oxidation state and passivation. **Scale bridging:** use MD only to calibrate transport or binding laws that influence mesoscopic behavior. **Evolution:** do not substitute steady-state FBA for origins-stage chemistry. **Model falsification:** compare alternatives (e.g., weak gradient, high dilution, absent sulfur, phosphorus limitation) and report negative regions of parameter space.

Suggested first repository structure: `models/geochemistry/`, `models/transport/`, `models/electrochemistry/`, `models/protometabolism/`, `models/compartments/`, `composites/`, `data/provenance/`, `experiments/`, `validation/`. Start with one integrated reproducible reference experiment before adding sophisticated model detail.